sight isn't necessary, and an AP should give you access up to 60-100 metres. Mehta affirms that, "Placement of the access point is critical… The thing to keep in mind is less obstruction and greater coverage area".

Typically, as the signal strength drops, so does the throughput, which is a rudimentary way of finding out what sort of 'spread' an AP has.

To a certain extent, government regulations have played their part in easing the transition to Wi-Fi. Currently, within a building or closed environment, a WLAN is de-licensed on the 2.4 GHz frequency spectrum. As far as IEEE networking standards go, the Indian government's Wireless Planning Commission (WPC) has permitted unlicensed usage of the IEEE 802.11b wireless networking protocol that allows for Wi-Fi connectivity at 11 Mbps.

Cisco's Network Consultant, Devendra M Kamtekar, says that the 802.11g standard regulations right now aren't very clear, and users may need clearance or licenses from the WPC for implementing the same—802.11g allows for a throughput of 54 Mbps. Both work on the 2.4 GHz frequency spectrum.

Vendors privately admit that 802.11g implementations within a closed environment are possible. However, before implementing the 802.11g standard, we strongly urge you to check the current regulations regarding this. At this point of time, outdoor usage in public areas requires licenses from the appropriate governing bodies such as the WPC.

ANAND MEHTA, Marketing Manager, D-Link india Ltd

> "The performance may not be the greatest, as compared to a traditional LAN…but **Wi-Fi** will let laptop users get **truly mobile** and **connect without wires**"

What do they think?

With their campus WLAN up and running, the Jains of Pathways are pretty upbeat about it. "The whole school, in a way, becomes a learning centre," says Prashant Jain. Students from standard VI upward carry laptops around the campus and in the classrooms. "Many a time, a thought or an idea comes to a person's mind, but by the time he gets back to the classroom or hostel room, the idea is lost. In a WLAN, this handicap is not at all there." This, as Jain points out, means that an idea is not limited to the classroom. Sify's clientele will agree with him—they can access the Internet within Bangalore's Java City chain of coffee bars, or the Casa Picola restaurant chain, or at any of the locations provided with hotspots.

Whether it's a chat with the boss as you sip coffee, or checking mail while snacking, wireless connectivity, in Appasamy's words, "Lets you get the job done, and get on with it". Their clientele is primarily drawn from business travellers who enter the city with Wi-Fi enabled laptops. Such globe-trotters love staying connected, and in IT-savvy Bangalore, Sify helps them do so by signing up with its service, right from the airport itself. Besides the good subscriber feedback, Appasamy goes on to state that Sify often gets enquiries from customers abroad who want to subscribe to the service before they even arrive in Bangalore.

Business travellers are also big Wi-Fi customers for LRM. Chandra says, "With 90 per cent of our business travellers wanting to stay connected, we have had more corporates booking the business centre, since there's wireless connectivity all over".

Wi-Fi certainly holds good for the retail segment as well. At Shoppers' Stop, Andheri, a Mumbai suburb, Anil Shankar says that with their Wi-Fi setup in place, they can change the store layout and decor, without having to worry about the networking infrastructure.

Problems... or are they?

Configuring the individual network nodes is pretty simple—in Windows, for instance, it's a matter of making the necessary changes in the network settings. Jain says that at Pathways, "The in-house systems administrator does this 2-minute job."

Service outages are an issue, though the people we spoke to didn't seem to be wrestling with too many of them. Chandra says, "LRM has yet to suffer any outages or network congestion." At present, their 11 Mbps Wi-Fi network is connected via a 512 Kbps pipe to their ISP. They were lucky—even the Blaster attack didn't clog their network. "We got the alert from the ISP," say Chandra and Giri, "and immediately contacted those with infected laptops". Their 24/7 technical support

Around the World in a Wi-Fi

Wi-Fi networks have risen from IEEE networking standards. The 802.11 IEEE covers the wireless networking standard. Commercial wireless access points were promoted in early 1995.

Wayport, a pioneering organisation in the Wi-Fi space, based in Austin, Texas, set up wireless networks in early 1997. The same year also saw the IEEE 802.11 standards being finalised.

Back in 1999, Apple offered Wi-Fi support along with the necessary hardware. 2001 saw the emergence of many companies that worked on public Wi-Fi, but several have since disappeared. In October 2002, the emergence of WPA, which has now replaced WEP, ensured better security. Intel worked on the Banias CPU architecture that led to the development of the Centrino platform in 2003. This platform works with a Pentium M CPU, the 855 chipset family, and a wireless network adapter that conforms to the 802.11b standard.

In the US, public Wi-Fi access has boomed. Starbucks, the coffee chain offers Wi-Fi services, as does McDonald's. Internationally, college campuses, hotels, airports and bookstores, such as the Barnes and Noble bookstore chain, are getting Wi-Fi enabled as well. There's a large laptop population in the student and white-collar community who actively use these services, both for work as well as leisure.